

A proof of the conjectured order of the recurrence for OEIS A281031

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1 The conjecture

The Online Encyclopedia of Integer Sequences records [A281031](#) as

Number of $n \times 5$ 0..1 arrays with no element equal to more than three of its king-move neighbors, with the exception of exactly one element, and with new values introduced in order 0 sequentially upwards

and carries in its Formula section the line:

`Empirical recurrence of order 98 (see link above)`

That line carries no separate signature; the entry itself is by R. H. Hardin (Jan 13 2017).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 4, last modified Jan 13 2017, the entry records no proof and links to none, so the statement is open. This note settles it.

The entry states only that the sequence satisfies a linear recurrence of order 98 and refers to a linked file for its coefficients, which this note does not use. What is proved below is the corresponding exact statement: the least order of a linear recurrence the sequence eventually satisfies, and the exact index beyond which it holds.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 5$ array over $\{0, \dots, 1\}$ with $n' = n$ rows.

The entry's neighbours, *king-move*, are the cells at the offsets

$$\mathcal{N} = \{(-1, -1), (-1, 0), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 0), (1, 1)\}$$

that lie inside the array, and $c(i, j)$ is the number of them carrying the same value as (i, j) .

Call a cell *bad* when $c(i, j) > 3$.

The condition is that exactly 1 cells are bad.

The entry's further clause says that reading the array row by row the values $0, \dots, 1$ first occur in increasing order.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Every neighbour offset moves at most one row, so whether a cell of row i is bad is decided by rows $i - 1$, i and $i + 1$. Take as state the last two rows, the number of bad cells so far capped at 2, and a counter for how many values have been introduced. Appending a row decides the row before it, and the accepting condition decides the last one.

An array of n' rows is then exactly a walk of length n' from the empty state to an accepting one, so $a(n) = w^T M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 2348 states, 1145 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 167$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of that 167×167 matrix; the bound is computed, not assumed.

5 The decision procedure

Let $c_1, \dots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^D c_i a(n - i)$, let $q(t) = t^D - \sum_{i=1}^D c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m = \tilde{w}^T L^{m-1} q(L) \tilde{f} \quad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a , one gets $u_m = a(n) - \sum_{i=1}^D c_i a(n - i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^T L^{m-1} y$ with $y = q(L) \tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L , of degree 167: writing that polynomial as $t^{167} - \sum_{i=1}^{167} \lambda_i t^{167-i}$ gives $u_{m+167} = \sum_{i=1}^{167} \lambda_i u_{m+167-i}$. Hence 167 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1} y$. The matrix M is never formed.

6 Results

Theorem 1. *For A281031, the least order of a linear recurrence that a eventually satisfies is exactly 98, and that recurrence holds for every $n > 109$, the entry’s own figure.*

Proof. The count is C-finite of order at most $S = 167$, so the minimal annihilator of the tail $(a(n))_{n \geq M}$ is the same for every M beyond S : the nilpotent part of L is killed after S steps. Taking such an M and 340 consecutive terms from there, the Berlekamp–Massey algorithm over $\mathbb{Q}$ returns the minimal linear recurrence generating them; since the sequence satisfies one of order at most S , $2S$ terms determine it. Its order is 98. That recurrence was then put through the residual test of Section 5: the last nonvanishing residual is at $n = 109$, followed by more than $S = 167$ consecutive vanishing ones, so it holds for every larger n . No recurrence of smaller order can hold eventually, since the tail’s minimal annihilator is what the algorithm returned. $\square$

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry’s English.

- **The published terms.** The walk count reproduces all 22 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry’s stated offset (1) rather than fitted. The first of them are 0, 58, 2321, 9854, 44395, 184290, 722952, 2696692, ...
- **An independent enumeration.** For $n = 1, \dots, 3$ every one of the 2^{5n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry’s published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 167$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry [A281031](#), revision 4, last modified Jan 13 2017.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.

4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.